

MH1301 Discrete Mathematics

Tutorial 5

Academic Year 2025/2026, Semester 2

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Overview

This tutorial develops recurrence-based counting and linear recurrence solving, with emphasis on:

- recurrence relations for binary and ternary strings with adjacency restrictions;
- tiling recurrences obtained from last-piece and state-based decompositions;
- closed-form solutions of first-order and second-order homogeneous recurrences;
- characteristic-root methods, repeated-root cases, and alternative transform methods;
- recurrence modelling in applied contexts, including tilings and population/catch models.

Question map.

- Ex. 8.1.10: binary strings containing two consecutive 0's.
- Ex. 8.1.9 and Ex. 8.1.11: ternary strings under forbidden-adjacency conditions.
- Additional exercise: tilings of an $n \times 1$ hallway with red and blue tiles.
- Ex. 8.2.2(a)–(e): solving homogeneous linear recurrences.
- Ex. 8.2.5: tilings of a $2 \times n$ board using 1×2 and 2×2 tiles.
- Ex. 8.2.6: a recurrence model for yearly lobster catches.

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Question 1 (Ex. 8.1.10: Bit strings containing two consecutive 0's)

Problem

A binary string is a string containing only 0's and 1's.

- Find a recurrence relation for the number of bit strings of length n that contain two consecutive 0's.
- What are the initial conditions?
- Find the answer when $n = 4$.

Solution

Method 1: Short end-pattern count

Let

$$a_n = \#\{\text{bit strings of length } n \text{ containing } 00\}.$$

(a) **Recurrence relation** For $n \geq 3$, every bit string counted by a_n ends in exactly one of

$$1, \quad 10, \quad 00.$$

We count each case separately.

- **Ends in 1.** Then the first $n - 1$ bits must already contain 00. Number of such strings:

$$a_{n-1}.$$

- **Ends in 10.** Then the first $n - 2$ bits must already contain 00. Number of such strings:

$$a_{n-2}.$$

- **Ends in 00.** Then the first $n - 2$ bits may be arbitrary, since appending 00 automatically guarantees the required property. Number of such strings:

$$2^{n-2}.$$

These three cases are disjoint and exhaustive, so

$$a_n = a_{n-1} + a_{n-2} + 2^{n-2} \quad (n \geq 3).$$

(b) Initial conditions For $n = 1$, the bit strings are

$$0, 1,$$

and neither contains two consecutive 0's. Hence

$$\boxed{a_1 = 0.}$$

For $n = 2$, the bit strings are

$$00, 01, 10, 11,$$

and only 00 contains two consecutive 0's. Hence

$$\boxed{a_2 = 1.}$$

Thus the full recurrence data are

$$\boxed{a_n = a_{n-1} + a_{n-2} + 2^{n-2} \quad (n \geq 3), \quad a_1 = 0, \quad a_2 = 1.}$$

(c) Value when $n = 4$ Using the recurrence,

$$a_3 = a_2 + a_1 + 2^1 = 1 + 0 + 2 = 3,$$

and then

$$a_4 = a_3 + a_2 + 2^2 = 3 + 1 + 4 = 8.$$

Therefore

$$\boxed{a_4 = 8.}$$

Method 2: Classify by the ending pattern

Let a_n be the number of binary strings of length n that contain two consecutive 0's.

(a) Recurrence relation For $n \geq 3$, classify such a string according to its final bits.

Case 1: The last bit is 1. Then the first $n-1$ bits must already form a valid length- $(n-1)$ string containing two consecutive 0's. So this case contributes

$$a_{n-1}.$$

Case 2: The last two bits are 10. Then the first $n-2$ bits must already contain two consecutive 0's. So this case contributes

$$a_{n-2}.$$

Case 3: The last two bits are 00. Any binary string of length $n-2$ can be followed by 00, and the resulting length- n string certainly contains two consecutive 0's. So this case contributes

$$2^{n-2}.$$

Since these three cases are disjoint and exhaustive,

$$\boxed{a_n = a_{n-1} + a_{n-2} + 2^{n-2} \quad (n \geq 3).}$$

(b) Initial conditions For $n = 1$, the strings are 0, 1, and neither contains two consecutive 0's. Hence

$$\boxed{a_1 = 0.}$$

For $n = 2$, the strings are 00, 01, 10, 11, and only 00 works. Hence

$$\boxed{a_2 = 1.}$$

Therefore

$$\boxed{a_n = a_{n-1} + a_{n-2} + 2^{n-2} \quad (n \geq 3), \quad a_1 = 0, \quad a_2 = 1.}$$

(c) Value when $n = 4$ First,

$$a_3 = a_2 + a_1 + 2^1 = 1 + 0 + 2 = 3.$$

Then,

$$a_4 = a_3 + a_2 + 2^2 = 3 + 1 + 4 = 8.$$

Therefore

$$\boxed{a_4 = 8.}$$

Method 3: Count by complement and then convert back

Let

$$b_n = \#\{\text{bit strings of length } n \text{ with no consecutive 0's}\}.$$

Then

$$a_n = 2^n - b_n.$$

(a) Recurrence relation We first find a recurrence for b_n .

A valid length- n string counted by b_n has one of the following forms:

- **Ends in 1.** Then the first $n - 1$ bits may be any valid string counted by b_{n-1} .
- **Ends in 0.** Then the preceding bit must be 1, so the string ends in 10. The first $n - 2$ bits may be any valid string counted by b_{n-2} .

Hence

$$b_n = b_{n-1} + b_{n-2} \quad (n \geq 3).$$

Now convert back to a_n :

$$a_n = 2^n - b_n = 2^n - (b_{n-1} + b_{n-2}).$$

Using

$$b_{n-1} = 2^{n-1} - a_{n-1}, \quad b_{n-2} = 2^{n-2} - a_{n-2},$$

we get

$$a_n = 2^n - ((2^{n-1} - a_{n-1}) + (2^{n-2} - a_{n-2})).$$

Thus

$$a_n = 2^n - 2^{n-1} - 2^{n-2} + a_{n-1} + a_{n-2} = 2^{n-2} + a_{n-1} + a_{n-2}.$$

Therefore

$$\boxed{a_n = a_{n-1} + a_{n-2} + 2^{n-2} \quad (n \geq 3).}$$

(b) Initial conditions For b_n , we have

$$b_1 = 2 \quad (0, 1), \quad b_2 = 3 \quad (01, 10, 11).$$

So

$$a_1 = 2^1 - b_1 = 2 - 2 = 0, \quad a_2 = 2^2 - b_2 = 4 - 3 = 1.$$

Hence

$$\boxed{a_1 = 0, \quad a_2 = 1.}$$

Therefore the full recurrence data are

$$\boxed{a_n = a_{n-1} + a_{n-2} + 2^{n-2} \quad (n \geq 3), \quad a_1 = 0, \quad a_2 = 1.}$$

(c) **Value when $n = 4$** Compute

$$b_3 = b_2 + b_1 = 3 + 2 = 5, \quad b_4 = b_3 + b_2 = 5 + 3 = 8.$$

Hence

$$a_4 = 2^4 - b_4 = 16 - 8 = 8.$$

Therefore

$$\boxed{a_4 = 8.}$$

Question 2 (Ex. 8.1.9: Ternary strings with no consecutive 0's)

Problem

A ternary string is a string containing only 0's, 1's, and 2's.

- Find a recurrence relation for the number of ternary strings of length n that do not contain two consecutive 0's.
- What are the initial conditions?
- Find the answer when $n = 4$.

Solution

Method 1: Short end-symbol count

Let

$$a_n = \#\{\text{ternary strings of length } n \text{ with no } 00\}.$$

For $n \geq 3$, split by the last symbol:

$$0, \quad 1, \quad 2.$$

If the last symbol is 0, the previous symbol must be 1 or 2, so this contributes

$$2a_{n-2}.$$

If the last symbol is 1 or 2, append it to any valid length- $(n-1)$ string, contributing

$$a_{n-1} + a_{n-1} = 2a_{n-1}.$$

Hence

$$a_n = 2a_{n-1} + 2a_{n-2}, \quad n \geq 3.$$

Also,

$$a_1 = 3, \quad a_2 = 9 - 1 = 8.$$

Thus

$$\boxed{a_n = 2a_{n-1} + 2a_{n-2} \quad (n \geq 3), \quad a_1 = 3, \quad a_2 = 8.}$$

Then

$$a_3 = 2(8) + 2(3) = 22, \quad a_4 = 2(22) + 2(8) = 60.$$

Hence

$$\boxed{a_4 = 60.}$$

Method 2: Classify by the last symbol

Let a_n be the number of ternary strings of length n that do not contain two consecutive 0's.

For $n \geq 3$, consider the last symbol.

Case 1: The last symbol is 0. Then the penultimate symbol cannot be 0; it must be 1 or 2. Thus the last two symbols are either 10 or 20. In either subcase, the first $n - 2$ symbols may be any valid ternary string counted by a_{n-2} . Hence this case contributes

$$2a_{n-2}.$$

Case 2: The last symbol is 1. Then the first $n - 1$ symbols may be any valid ternary string counted by a_{n-1} .

Case 3: The last symbol is 2. Exactly as in Case 2, this contributes another

$$a_{n-1}.$$

Adding the three disjoint cases gives

$$a_n = 2a_{n-1} + 2a_{n-2}, \quad n \geq 3.$$

Now compute the initial conditions. For $n = 1$, all three strings

$$0, 1, 2$$

are valid, so

$$a_1 = 3.$$

For $n = 2$, there are $3^2 = 9$ ternary strings in total, and only 00 is forbidden. Hence

$$a_2 = 9 - 1 = 8.$$

Therefore

$$\boxed{a_n = 2a_{n-1} + 2a_{n-2} \quad (n \geq 3), \quad a_1 = 3, \quad a_2 = 8.}$$

For part (c),

$$a_3 = 2a_2 + 2a_1 = 2(8) + 2(3) = 22,$$

and then

$$a_4 = 2a_3 + 2a_2 = 2(22) + 2(8) = 60.$$

Thus

$$\boxed{a_4 = 60.}$$

Method 3: Use end-state variables and eliminate them

Let

$$z_n = \#\{\text{valid length-}n \text{ strings ending in } 0\},$$

and

$$u_n = \#\{\text{valid length-}n \text{ strings ending in } 1 \text{ or } 2\}.$$

Then

$$a_n = z_n + u_n.$$

We derive recurrences for z_n and u_n .

To end in 0, the previous symbol must be either 1 or 2. Therefore

$$z_n = u_{n-1}.$$

To end in 1 or 2, take any valid length- $(n-1)$ string and append either 1 or 2. This gives

$$u_n = 2a_{n-1}.$$

Hence

$$a_n = z_n + u_n = u_{n-1} + 2a_{n-1}.$$

Using $u_{n-1} = 2a_{n-2}$, we obtain

$$a_n = 2a_{n-1} + 2a_{n-2}, \quad n \geq 3.$$

Now compute the initial conditions:

$$a_1 = 3$$

because 0, 1, 2 are all valid, and

$$a_2 = 8$$

because the only invalid length-2 ternary string is 00.

Thus

$$\boxed{a_n = 2a_{n-1} + 2a_{n-2} \quad (n \geq 3), \quad a_1 = 3, \quad a_2 = 8.}$$

Finally,

$$a_3 = 2(8) + 2(3) = 22, \quad a_4 = 2(22) + 2(8) = 60,$$

so

$$\boxed{a_4 = 60.}$$

Question 3 (Ex. 8.1.11: Ternary strings with no consecutive 0's or 1's)

Problem

- (a) Find a recurrence relation for the number of ternary strings of length n that do not contain two consecutive 0's or two consecutive 1's.
- (b) What are the initial conditions?
- (c) Find the answer when $n = 4$.

Solution

Method 1: Short end-symbol count

Let

$$a_n = \#\{\text{valid ternary strings of length } n\}.$$

The last symbol is 0, 1, or 2. If it is 0, the previous symbol must be 1 or 2, so this contributes

$$a_{1,n-1} + a_{2,n-1} = a_{n-1} - a_{0,n-1}.$$

Similarly, ending in 1 contributes

$$a_{0,n-1} + a_{2,n-1} = a_{n-1} - a_{1,n-1}.$$

Ending in 2 contributes

$$a_{n-1}.$$

Using symmetry $a_{0,n-1} = a_{1,n-1}$ and $a_{2,n-1} = a_{n-2}$, one gets

$$a_n = 2a_{n-1} + a_{n-2}, \quad n \geq 3.$$

Equivalently, a valid length- n string is obtained by appending:

- 2 to any valid length- $(n-1)$ string $\Rightarrow a_{n-1}$;
- 0 or 1 to a valid length- $(n-1)$ string ending in a different symbol, giving altogether $a_{n-1} + a_{n-2}$.

Hence

$$a_n = 2a_{n-1} + a_{n-2}.$$

Also,

$$a_1 = 3, \quad a_2 = 9 - 2 = 7.$$

Thus

$$a_n = 2a_{n-1} + a_{n-2} \quad (n \geq 3), \quad a_1 = 3, \quad a_2 = 7.$$

Then

$$a_3 = 2(7) + 3 = 17, \quad a_4 = 2(17) + 7 = 41.$$

Hence

$$a_4 = 41.$$

Method 2: Three end-state counts

Let a_n be the number of ternary strings of length n containing neither 00 nor 11.

Define

$$a_{i,n} = \#\{\text{valid length-}n \text{ strings ending in } i\}, \quad i \in \{0, 1, 2\}.$$

Then

$$a_n = a_{0,n} + a_{1,n} + a_{2,n}.$$

From the adjacency restrictions:

$$a_{0,n} = a_{1,n-1} + a_{2,n-1}, \quad a_{1,n} = a_{0,n-1} + a_{2,n-1}, \quad a_{2,n} = a_{n-1}.$$

Summing gives

$$a_n = (a_{1,n-1} + a_{2,n-1}) + (a_{0,n-1} + a_{2,n-1}) + a_{n-1} = 2a_{n-1} + a_{2,n-1}.$$

Since a string ending in 2 at length $n-1$ corresponds to any valid string of length $n-2$ followed by 2,

$$a_{2,n-1} = a_{n-2}.$$

Hence

$$\boxed{a_n = 2a_{n-1} + a_{n-2}}, \quad n \geq 3.$$

For $n = 1$:

$$a_1 = 3 \quad (0, 1, 2).$$

For $n = 2$:

$$3^2 = 9 \text{ strings, forbidden: } 00, 11,$$

so

$$a_2 = 7.$$

Thus

$$\boxed{a_n = 2a_{n-1} + a_{n-2} \ (n \geq 3), \quad a_1 = 3, \ a_2 = 7.}$$

$$a_3 = 2a_2 + a_1 = 2(7) + 3 = 17,$$

$$a_4 = 2a_3 + a_2 = 2(17) + 7 = 41.$$

$$\boxed{a_4 = 41.}$$

Method 3: Separate the strings ending in 2 from those ending in 0 or 1

Let

$$b_n = \#\{\text{valid length-}n \text{ strings ending in } 2\},$$

$$c_n = \#\{\text{valid length-}n \text{ strings ending in } 0 \text{ or } 1\}.$$

Then

$$a_n = b_n + c_n.$$

We first express b_n : to obtain a valid length- n string ending in 2, we may append 2 to any valid length- $(n-1)$ string. Hence

$$b_n = a_{n-1}.$$

Next we count c_n . A valid length- n string ending in 0 or 1 can arise in two disjoint ways:

- append 0 or 1 to a valid length- $(n-1)$ string ending in 2; this gives

$$2b_{n-1}$$

possibilities;

- append the *opposite* symbol to a valid length- $(n-1)$ string ending in 0 or 1; for each such string there is exactly one allowed choice, so this contributes

$$c_{n-1}.$$

Therefore

$$c_n = 2b_{n-1} + c_{n-1}.$$

Substitute

$$b_{n-1} = a_{n-2}, \quad c_{n-1} = a_{n-1} - b_{n-1} = a_{n-1} - a_{n-2},$$

to obtain

$$c_n = 2a_{n-2} + (a_{n-1} - a_{n-2}) = a_{n-1} + a_{n-2}.$$

Hence

$$a_n = b_n + c_n = a_{n-1} + (a_{n-1} + a_{n-2}) = 2a_{n-1} + a_{n-2}.$$

The initial conditions are again

$$a_1 = 3, \quad a_2 = 7.$$

Thus

$$\boxed{a_n = 2a_{n-1} + a_{n-2} \quad (n \geq 3), \quad a_1 = 3, \quad a_2 = 7.}$$

Finally,

$$a_4 = 2a_3 + a_2 = 2(17) + 7 = 41,$$

so

$$\boxed{a_4 = 41.}$$

Question 4 (Additional Exercise: Tiling an $n \times 1$ hallway)

Problem

You need to tile an $n \times 1$ hallway with an unlimited supply of 1×1 red tiles, 2×1 red tiles, and 2×1 blue tiles. Write down the recurrence formula and initial conditions for the number of ways to tile:

- (a) an $n \times 1$ hallway for which you are allowed to use all types of tiles;
- (b) an $n \times 1$ hallway for which you are allowed to use all types of tiles, but no two red tiles are consecutive.

Solution

Method 1: Short last-tile count

- (a) Let f_n be the number of tilings. The last tile is

$$\text{red } 1, \quad \text{red } 2, \quad \text{blue } 2.$$

Hence

$$f_n = f_{n-1} + 2f_{n-2}, \quad n \geq 3.$$

Also,

$$f_1 = 1, \quad f_2 = 3.$$

Thus

$$\boxed{f_n = f_{n-1} + 2f_{n-2} \quad (n \geq 3), \quad f_1 = 1, \quad f_2 = 3.}$$

- (b) Let g_n be the number of valid tilings with no two red tiles consecutive. The ending must be one of

$$\text{blue } 2, \quad \text{blue } 2 + \text{red } 1, \quad \text{blue } 2 + \text{red } 2,$$

so

$$g_n = g_{n-2} + g_{n-3} + g_{n-4}, \quad n \geq 5.$$

Also,

$$g_1 = 1, \quad g_2 = 2, \quad g_3 = 2, \quad g_4 = 4.$$

Thus

$$\boxed{g_n = g_{n-2} + g_{n-3} + g_{n-4} \quad (n \geq 5), \quad g_1 = 1, \quad g_2 = 2, \quad g_3 = 2, \quad g_4 = 4.}$$

Method 2: Classify by the last tile

(a) Let f_n be the number of tilings of an $n \times 1$ hallway using all three tile types.

For $n \geq 3$, consider the last tile.

- If the last tile is a 1×1 red tile, then the first $n - 1$ positions may be tiled in f_{n-1} ways.
- If the last tile is a 2×1 red tile, then the first $n - 2$ positions may be tiled in f_{n-2} ways.
- If the last tile is a 2×1 blue tile, then the first $n - 2$ positions may again be tiled in f_{n-2} ways.

Hence

$$f_n = f_{n-1} + 2f_{n-2}, \quad n \geq 3.$$

Initial conditions:

$$f_1 = 1$$

because only the 1×1 red tile fits, and

$$f_2 = 3$$

because the valid tilings are

$$(\text{red } 1) + (\text{red } 1), \quad (\text{red } 2), \quad (\text{blue } 2).$$

Therefore

$$f_n = f_{n-1} + 2f_{n-2} \quad (n \geq 3), \quad f_1 = 1, \quad f_2 = 3.$$

(b) Let g_n be the number of tilings of an $n \times 1$ hallway in which no two red tiles are consecutive.

For $n \geq 5$, classify by the last tile.

If the last tile is a 2×1 blue tile, then the first $n - 2$ positions may be tiled by any valid tiling counted by g_{n-2} .

If the last tile is a 1×1 red tile, then the preceding tile cannot be red, so it must be a 2×1 blue tile. Thus the tiling ends in

$$(\text{blue } 2) + (\text{red } 1),$$

and the first $n - 3$ positions may be tiled in g_{n-3} ways.

If the last tile is a 2×1 red tile, then the preceding tile must again be a 2×1 blue tile, so the tiling ends in

$$(\text{blue } 2) + (\text{red } 2),$$

and the first $n - 4$ positions may be tiled in g_{n-4} ways.

Hence

$$g_n = g_{n-2} + g_{n-3} + g_{n-4}, \quad n \geq 5.$$

Now compute the initial conditions explicitly:

$$g_1 = 1$$

(the only tiling is a single 1×1 red tile);

$$g_2 = 2$$

(the tilings are red 2 and blue 2; two red 1×1 tiles are forbidden);

$$g_3 = 2$$

(the tilings are blue 2+red 1, and red 1+blue 2);

$$g_4 = 4$$

(the tilings are blue 2+blue 2, blue 2+red 2, red 2+blue 2, red 1+blue 2+red 1).

Therefore

$g_n = g_{n-2} + g_{n-3} + g_{n-4} \quad (n \geq 5), \quad g_1 = 1, \quad g_2 = 2, \quad g_3 = 2, \quad g_4 = 4.$

Method 3: State decomposition

(a) For the unrestricted problem, let

$$F_n = \#\{\text{valid tilings of length } n\}.$$

A valid tiling is obtained by choosing its *first* tile instead of its last tile:

- first tile 1×1 red: F_{n-1} possibilities remain;
- first tile 2×1 red: F_{n-2} possibilities remain;
- first tile 2×1 blue: F_{n-2} possibilities remain.

Thus

$$F_n = F_{n-1} + 2F_{n-2}, \quad n \geq 3,$$

with

$$F_1 = 1, \quad F_2 = 3.$$

So

$$\boxed{F_n = F_{n-1} + 2F_{n-2} \quad (n \geq 3), \quad F_1 = 1, \quad F_2 = 3.}$$

(b) For the restricted problem, let

$$B_n = \#\{\text{valid length-}n \text{ tilings ending in a blue } 2 \times 1 \text{ tile}\},$$

$$R_n = \#\{\text{valid length-}n \text{ tilings ending in a red tile}\}.$$

Then

$$g_n = B_n + R_n.$$

Now derive recurrences:

- To end in a blue 2×1 tile, remove that last tile. What remains is any valid tiling of length $n - 2$. Hence

$$B_n = g_{n-2}.$$

- To end in a red 1×1 tile, the preceding tile must be blue, so the count is B_{n-1} .
- To end in a red 2×1 tile, the preceding tile must also be blue, so the count is B_{n-2} .

Therefore

$$R_n = B_{n-1} + B_{n-2} = g_{n-3} + g_{n-4}.$$

Hence

$$g_n = B_n + R_n = g_{n-2} + g_{n-3} + g_{n-4}, \quad n \geq 5.$$

With the same direct initial counts

$$g_1 = 1, \quad g_2 = 2, \quad g_3 = 2, \quad g_4 = 4,$$

we obtain

$$\boxed{g_n = g_{n-2} + g_{n-3} + g_{n-4} \quad (n \geq 5), \quad g_1 = 1, \quad g_2 = 2, \quad g_3 = 2, \quad g_4 = 4.}$$

Question 5 (Ex. 8.2.2(a)–(e): Solve the homogeneous recurrence relations)

Problem

Solve the following homogeneous recurrence relations.

- (a) $a_n = 2a_{n-1}$ for $n \geq 1$, $a_0 = 3$.
- (b) $a_n = a_{n-1}$ for $n \geq 1$, $a_0 = 2$.
- (c) $a_n = 5a_{n-1} - 6a_{n-2}$ for $n \geq 2$, $a_0 = 1$, $a_1 = 0$.
- (d) $a_n = 4a_{n-1} - 4a_{n-2}$ for $n \geq 2$, $a_0 = 6$, $a_1 = 8$.
- (e) $a_n = -4a_{n-1} - 4a_{n-2}$ for $n \geq 2$, $a_0 = 0$, $a_1 = 1$.

Solution

Method 1: Characteristic equations

- (a) The recurrence

$$a_n = 2a_{n-1}$$

has characteristic equation

$$r = 2.$$

Hence the general form is

$$a_n = h 2^n.$$

Use $a_0 = 3$:

$$3 = a_0 = h 2^0 = h.$$

So

$$\boxed{a_n = 3 \cdot 2^n.}$$

- (b) The recurrence

$$a_n = a_{n-1}$$

has characteristic equation

$$r = 1.$$

Hence

$$a_n = h 1^n = h.$$

Using $a_0 = 2$, we get $h = 2$. Therefore

$$\boxed{a_n = 2.}$$

(c) The characteristic equation is

$$r^2 - 5r + 6 = 0.$$

Factor:

$$r^2 - 5r + 6 = (r - 2)(r - 3) = 0.$$

Thus the roots are 2 and 3, so

$$a_n = h_1 2^n + h_2 3^n.$$

Apply the initial conditions:

$$a_0 = 1 \implies h_1 + h_2 = 1,$$

$$a_1 = 0 \implies 2h_1 + 3h_2 = 0.$$

Solving,

$$h_1 = 3, \quad h_2 = -2.$$

Therefore

$$\boxed{a_n = 3 \cdot 2^n - 2 \cdot 3^n.}$$

(d) The characteristic equation is

$$r^2 - 4r + 4 = 0 = (r - 2)^2.$$

There is a repeated root $r = 2$, so the general form is

$$a_n = (h_1 + h_2 n) 2^n.$$

Use $a_0 = 6$:

$$6 = (h_1 + 0) 2^0 \implies h_1 = 6.$$

Use $a_1 = 8$:

$$8 = (6 + h_2) 2.$$

Hence

$$6 + h_2 = 4 \implies h_2 = -2.$$

So

$$a_n = (6 - 2n) 2^n.$$

Equivalently,

$$a_n = 6 \cdot 2^n - n \cdot 2^{n+1}.$$

Thus

$$\boxed{a_n = (6 - 2n) 2^n.}$$

(e) The characteristic equation is

$$r^2 + 4r + 4 = 0 = (r + 2)^2.$$

There is a repeated root $r = -2$, so

$$a_n = (h_1 + h_2 n)(-2)^n.$$

Use $a_0 = 0$:

$$0 = (h_1 + 0)(-2)^0 \implies h_1 = 0.$$

Use $a_1 = 1$:

$$1 = (0 + h_2)(-2) \implies h_2 = -\frac{1}{2}.$$

Therefore

$$a_n = -\frac{1}{2} n(-2)^n = n(-2)^{n-1}.$$

Hence

$$\boxed{a_n = n(-2)^{n-1}.$$

Method 2: Characteristic-root formulas

(a)

$$a_n = 2a_{n-1}.$$

$$a_n = C2^n.$$

$$\text{Using } a_0 = 3 \Rightarrow C = 3,$$

$$\boxed{a_n = 3 \cdot 2^n.}$$

(b)

$$a_n = a_{n-1}.$$

$$a_n = C.$$

$$\text{Using } a_0 = 2 \Rightarrow C = 2,$$

$$\boxed{a_n = 2.}$$

(c)

$$r^2 - 5r + 6 = 0 = (r - 2)(r - 3).$$

$$a_n = C_1 2^n + C_2 3^n.$$

$$\text{Using } a_0 = 1, a_1 = 0:$$

$$C_1 + C_2 = 1, \quad 2C_1 + 3C_2 = 0 \Rightarrow C_1 = 3, C_2 = -2.$$

$$\boxed{a_n = 3 \cdot 2^n - 2 \cdot 3^n.}$$

(d)

$$r^2 - 4r + 4 = 0 = (r - 2)^2.$$

$$a_n = (C_1 + C_2 n)2^n.$$

$$\text{Using } a_0 = 6, a_1 = 8:$$

$$C_1 = 6, \quad 2(6 + C_2) = 8 \Rightarrow C_2 = -2.$$

$$\boxed{a_n = (6 - 2n)2^n.}$$

(e)

$$r^2 + 4r + 4 = 0 = (r + 2)^2.$$

$$a_n = (C_1 + C_2 n)(-2)^n.$$

$$\text{Using } a_0 = 0, a_1 = 1:$$

$$C_1 = 0, \quad -2C_2 = 1 \Rightarrow C_2 = -\frac{1}{2}.$$

$$a_n = -\frac{1}{2}n(-2)^n = n(-2)^{n-1}.$$

$$\boxed{a_n = n(-2)^{n-1}.}$$

Method 3: Factor the recurrence stepwise into first-order recurrences

(a) Starting from

$$a_n = 2a_{n-1},$$

divide by 2^n :

$$\frac{a_n}{2^n} = \frac{a_{n-1}}{2^{n-1}}.$$

Hence the sequence $\frac{a_n}{2^n}$ is constant. Let that constant be C . Then

$$a_n = C2^n.$$

Using $a_0 = 3$, we obtain $C = 3$. Therefore

$$\boxed{a_n = 3 \cdot 2^n.}$$

(b) From

$$a_n = a_{n-1},$$

we immediately obtain

$$a_n = a_{n-1} = a_{n-2} = \cdots = a_0 = 2.$$

Hence

$$\boxed{a_n = 2.}$$

(c) Rewrite

$$a_n - 2a_{n-1} = 3(a_{n-1} - 2a_{n-2}).$$

Define

$$b_n = a_n - 2a_{n-1}.$$

Then

$$b_n = 3b_{n-1}.$$

Also,

$$b_1 = a_1 - 2a_0 = 0 - 2(1) = -2.$$

Therefore

$$b_n = -2 \cdot 3^{n-1}.$$

So a_n satisfies the first-order nonhomogeneous recurrence

$$a_n - 2a_{n-1} = -2 \cdot 3^{n-1}.$$

Solve it by writing

$$a_n = c2^n + A3^n.$$

Substitute into $a_n - 2a_{n-1} = -2 \cdot 3^{n-1}$:

$$(c2^n + A3^n) - 2(c2^{n-1} + A3^{n-1}) = A3^{n-1} = -2 \cdot 3^{n-1}.$$

Hence

$$A = -2.$$

Now use $a_0 = 1$:

$$1 = c - 2 \implies c = 3.$$

Therefore

$$\boxed{a_n = 3 \cdot 2^n - 2 \cdot 3^n.}$$

(d) Rewrite

$$a_n - 2a_{n-1} = 2(a_{n-1} - 2a_{n-2}).$$

Define

$$b_n = a_n - 2a_{n-1}.$$

Then

$$b_n = 2b_{n-1}.$$

Also,

$$b_1 = a_1 - 2a_0 = 8 - 12 = -4.$$

Hence

$$b_n = -4 \cdot 2^{n-1} = -2^{n+1}.$$

So a_n satisfies

$$a_n - 2a_{n-1} = -2^{n+1}.$$

Because the homogeneous part has root 2, try a particular solution of the form

$$a_n^{(p)} = An2^n.$$

Then

$$a_n^{(p)} - 2a_{n-1}^{(p)} = An2^n - 2A(n-1)2^{n-1} = A2^n.$$

We need this to equal -2^{n+1} , so

$$A = -2.$$

Thus the general solution is

$$a_n = c2^n - 2n2^n = (c - 2n)2^n.$$

Use $a_0 = 6$, giving $c = 6$. Therefore

$$\boxed{a_n = (6 - 2n)2^n.}$$

(e) Rewrite

$$a_n + 2a_{n-1} = -2(a_{n-1} + 2a_{n-2}).$$

Define

$$b_n = a_n + 2a_{n-1}.$$

Then

$$b_n = -2b_{n-1}.$$

Also,

$$b_1 = a_1 + 2a_0 = 1 + 0 = 1.$$

Hence

$$b_n = (-2)^{n-1}.$$

So

$$a_n + 2a_{n-1} = (-2)^{n-1}.$$

Because the homogeneous part has root -2 , try a particular solution

$$a_n^{(p)} = An(-2)^n.$$

Then

$$a_n^{(p)} + 2a_{n-1}^{(p)} = An(-2)^n + 2A(n-1)(-2)^{n-1} = An(-2)^n - A(n-1)(-2)^n = A(-2)^n.$$

We want this to equal $(-2)^{n-1}$, so

$$A = -\frac{1}{2}.$$

Thus

$$a_n = c(-2)^n - \frac{1}{2}n(-2)^n.$$

Use $a_0 = 0$, giving $c = 0$. Therefore

$$a_n = n(-2)^{n-1}.$$

Hence

$$\boxed{a_n = n(-2)^{n-1}.$$

Question 6 (Ex. 8.2.5: Tiling a $2 \times n$ board)

Problem

Let a_n be the number of ways to tile a $2 \times n$ rectangular board using 1×2 and 2×2 pieces.

- (a) Calculate the values a_1 and a_2 .
- (b) Find a recurrence relation for the sequence (a_n) .
- (c) Solve the recurrence relation in part (b).

Solution

Method 1: Short decomposition + closed form

(a)

$$a_1 = 1, \quad a_2 = 3.$$

Hence

$$\boxed{a_1 = 1, \quad a_2 = 3.}$$

(b) The last part is one of:

vertical 1×2 , square 2×2 , two horizontal 1×2 .

Hence

$$a_n = a_{n-1} + 2a_{n-2}, \quad n \geq 3.$$

Thus

$$\boxed{a_n = a_{n-1} + 2a_{n-2} \quad (n \geq 3), \quad a_1 = 1, \quad a_2 = 3.}$$

(c)

$$r^2 - r - 2 = 0 = (r - 2)(r + 1),$$

so

$$a_n = A2^n + B(-1)^n.$$

Using

$$a_1 = 1, \quad a_2 = 3,$$

we get

$$2A - B = 1, \quad 4A + B = 3.$$

Hence

$$A = \frac{2}{3}, \quad B = \frac{1}{3}.$$

Therefore

$$\boxed{a_n = \frac{2}{3} 2^n + \frac{1}{3} (-1)^n = \frac{2^{n+1} + (-1)^n}{3}.$$

Method 2: Last-column decomposition and characteristic roots

(a) For a 2×1 board, the only possible tiling is one vertical 1×2 domino. Hence

$$a_1 = 1.$$

For a 2×2 board, the possible tilings are:

- two vertical 1×2 dominoes;
- two horizontal 1×2 dominoes;
- one 2×2 square.

Therefore

$$a_2 = 3.$$

Thus

$$\boxed{a_1 = 1, \quad a_2 = 3.}$$

(b) For $n \geq 3$, examine the rightmost part of a tiling of the $2 \times n$ board.

- If the last column is covered by one vertical 1×2 domino, then the remaining board is $2 \times (n - 1)$, giving a_{n-1} possibilities.
- If the last two columns are covered by one 2×2 square, then the remaining board is $2 \times (n - 2)$, giving a_{n-2} possibilities.
- If the last two columns are covered by two horizontal 1×2 dominoes, then again the remaining board is $2 \times (n - 2)$, giving another a_{n-2} possibilities.

These cases are disjoint and exhaustive, so

$$a_n = a_{n-1} + 2a_{n-2}, \quad n \geq 3.$$

Hence

$$\boxed{a_n = a_{n-1} + 2a_{n-2} \quad (n \geq 3), \quad a_1 = 1, \quad a_2 = 3.}$$

(c) Solve

$$a_n = a_{n-1} + 2a_{n-2}.$$

The characteristic equation is

$$r^2 - r - 2 = 0,$$

which factors as

$$(r - 2)(r + 1) = 0.$$

Hence

$$a_n = h_1 2^n + h_2 (-1)^n.$$

Use $a_1 = 1$ and $a_2 = 3$:

$$1 = 2h_1 - h_2,$$

$$3 = 4h_1 + h_2.$$

Adding these equations gives

$$4 = 6h_1 \implies h_1 = \frac{2}{3}.$$

Then

$$h_2 = 2h_1 - 1 = \frac{4}{3} - 1 = \frac{1}{3}.$$

Therefore

$$a_n = \frac{2}{3} 2^n + \frac{1}{3} (-1)^n.$$

Equivalently,

$$a_n = \frac{2^{n+1} + (-1)^n}{3}.$$

Thus

$$a_n = \frac{2}{3} 2^n + \frac{1}{3} (-1)^n = \frac{2^{n+1} + (-1)^n}{3}.$$

Method 3: Use a_0 and solve by generating functions

(a) It is convenient to define

$$a_0 = 1,$$

corresponding to the empty tiling of a 2×0 board. Then, as above,

$$a_1 = 1, \quad a_2 = 3.$$

So

$$a_1 = 1, \quad a_2 = 3.$$

(b) Using the same rightmost-piece analysis as in Method 2, we obtain

$$a_n = a_{n-1} + 2a_{n-2}, \quad n \geq 2,$$

if we use the convention $a_0 = 1$. This is equivalent to the $n \geq 3$ version together with $a_1 = 1, a_2 = 3$. Hence

$$a_n = a_{n-1} + 2a_{n-2} \quad (n \geq 2), \quad a_0 = 1, \quad a_1 = 1.$$

(c) Let

$$A(x) = \sum_{n \geq 0} a_n x^n.$$

From

$$a_n - a_{n-1} - 2a_{n-2} = 0 \quad (n \geq 2),$$

multiply by x^n and sum for $n \geq 2$:

$$\sum_{n \geq 2} a_n x^n - \sum_{n \geq 2} a_{n-1} x^n - 2 \sum_{n \geq 2} a_{n-2} x^n = 0.$$

Rewrite each sum:

$$\sum_{n \geq 2} a_n x^n = A(x) - a_0 - a_1 x = A(x) - 1 - x,$$

$$\sum_{n \geq 2} a_{n-1} x^n = x \sum_{n \geq 2} a_{n-1} x^{n-1} = x(A(x) - a_0) = x(A(x) - 1),$$

$$\sum_{n \geq 2} a_{n-2} x^n = x^2 A(x).$$

Therefore

$$A(x) - 1 - x - x(A(x) - 1) - 2x^2 A(x) = 0.$$

Simplifying,

$$A(x)(1 - x - 2x^2) = 1.$$

Hence

$$A(x) = \frac{1}{1 - x - 2x^2} = \frac{1}{(1 - 2x)(1 + x)}.$$

Use partial fractions:

$$\frac{1}{(1 - 2x)(1 + x)} = \frac{2/3}{1 - 2x} + \frac{1/3}{1 + x}.$$

Therefore

$$A(x) = \frac{2}{3} \sum_{n \geq 0} (2x)^n + \frac{1}{3} \sum_{n \geq 0} (-x)^n = \sum_{n \geq 0} \left(\frac{2}{3} 2^n + \frac{1}{3} (-1)^n \right) x^n.$$

So

$$a_n = \frac{2}{3} 2^n + \frac{1}{3} (-1)^n.$$

Thus

$$a_n = \frac{2}{3} 2^n + \frac{1}{3} (-1)^n = \frac{2^{n+1} + (-1)^n}{3}.$$

Question 7 (Ex. 8.2.6: Lobster catches)

Problem

A model for the number of lobsters caught per year is based on the assumption that the number of lobsters caught in a year is the average of the number caught in the two previous years.

- Find a recurrence relation for the sequence (L_n) , where L_n is the number of lobsters caught in year n .
- Solve L_n if 100,000 lobsters were caught in year 1 and 300,000 lobsters were caught in year 2.

Solution

Method 1: Short recurrence + root form

(a)

$$L_n = \frac{L_{n-1} + L_{n-2}}{2}, \quad n \geq 3.$$

Hence

$$L_n = \frac{1}{2}L_{n-1} + \frac{1}{2}L_{n-2} \quad (n \geq 3).$$

(b)

$$r^2 - \frac{1}{2}r - \frac{1}{2} = 0 \iff 2r^2 - r - 1 = (2r + 1)(r - 1) = 0,$$

so

$$L_n = A + B \left(-\frac{1}{2}\right)^n.$$

Using

$$L_1 = 100000, \quad L_2 = 300000,$$

we get

$$A - \frac{1}{2}B = 100000, \quad A + \frac{1}{4}B = 300000.$$

Hence

$$B = \frac{800000}{3}, \quad A = \frac{700000}{3}.$$

Therefore

$$L_n = \frac{700000}{3} + \frac{800000}{3} \left(-\frac{1}{2}\right)^n.$$

Method 2: Characteristic equation

(a) Since L_n is the average of the two previous yearly catches,

$$L_n = \frac{L_{n-1} + L_{n-2}}{2}, \quad n \geq 3.$$

Thus

$$L_n = \frac{1}{2}L_{n-1} + \frac{1}{2}L_{n-2} \quad (n \geq 3).$$

(b) We solve

$$L_n = \frac{1}{2}L_{n-1} + \frac{1}{2}L_{n-2}, \quad L_1 = 100000, \quad L_2 = 300000.$$

The characteristic equation is

$$r^2 - \frac{1}{2}r - \frac{1}{2} = 0.$$

Multiply by 2:

$$2r^2 - r - 1 = 0.$$

Factor:

$$2r^2 - r - 1 = (2r + 1)(r - 1) = 0.$$

Hence the roots are

$$r_1 = 1, \quad r_2 = -\frac{1}{2}.$$

Therefore

$$L_n = A + B \left(-\frac{1}{2}\right)^n.$$

Use the initial conditions:

$$100000 = A - \frac{1}{2}B,$$

$$300000 = A + \frac{1}{4}B.$$

Subtract the first equation from the second:

$$200000 = \frac{1}{4}B + \frac{1}{2}B = \frac{3}{4}B.$$

Hence

$$B = \frac{800000}{3}.$$

Substitute back:

$$A = 100000 + \frac{1}{2} \cdot \frac{800000}{3} = 100000 + \frac{400000}{3} = \frac{700000}{3}.$$

Therefore

$$L_n = \frac{700000}{3} + \frac{800000}{3} \left(-\frac{1}{2}\right)^n.$$

Hence

$$L_n = \frac{700000}{3} + \frac{800000}{3} \left(-\frac{1}{2}\right)^n.$$

Method 3: Convert the model into a recurrence for first differences

(a) The recurrence relation is still

$$L_n = \frac{1}{2}L_{n-1} + \frac{1}{2}L_{n-2} \quad (n \geq 3).$$

(b) Define the first-difference sequence

$$D_n = L_n - L_{n-1} \quad (n \geq 2).$$

Then

$$\begin{aligned} D_n &= L_n - L_{n-1} \\ &= \frac{1}{2}(L_{n-1} + L_{n-2}) - L_{n-1} \\ &= \frac{1}{2}(L_{n-2} - L_{n-1}) \\ &= -\frac{1}{2}(L_{n-1} - L_{n-2}) \\ &= -\frac{1}{2}D_{n-1}. \end{aligned}$$

So the first differences form a geometric sequence with ratio $-\frac{1}{2}$.

Now

$$D_2 = L_2 - L_1 = 300000 - 100000 = 200000.$$

Hence

$$D_n = 200000 \left(-\frac{1}{2}\right)^{n-2}, \quad n \geq 2.$$

Now recover L_n by summing the differences:

$$L_n = L_1 + \sum_{j=2}^n D_j = 100000 + 200000 \sum_{j=2}^n \left(-\frac{1}{2}\right)^{j-2}.$$

Let $m = j - 2$. Then

$$L_n = 100000 + 200000 \sum_{m=0}^{n-2} \left(-\frac{1}{2}\right)^m.$$

This is a finite geometric series:

$$\sum_{m=0}^{n-2} r^m = \frac{1 - r^{n-1}}{1 - r}, \quad r = -\frac{1}{2}.$$

Therefore

$$\sum_{m=0}^{n-2} \left(-\frac{1}{2}\right)^m = \frac{1 - \left(-\frac{1}{2}\right)^{n-1}}{1 + \frac{1}{2}} = \frac{1 - \left(-\frac{1}{2}\right)^{n-1}}{3/2} = \frac{2}{3} \left(1 - \left(-\frac{1}{2}\right)^{n-1}\right).$$

So

$$\begin{aligned} L_n &= 100000 + 200000 \cdot \frac{2}{3} \left(1 - \left(-\frac{1}{2}\right)^{n-1}\right) \\ &= 100000 + \frac{400000}{3} - \frac{400000}{3} \left(-\frac{1}{2}\right)^{n-1} \\ &= \frac{700000}{3} - \frac{400000}{3} \left(-\frac{1}{2}\right)^{n-1}. \end{aligned}$$

Since

$$-\frac{400000}{3} \left(-\frac{1}{2}\right)^{n-1} = \frac{800000}{3} \left(-\frac{1}{2}\right)^n,$$

this becomes

$$L_n = \frac{700000}{3} + \frac{800000}{3} \left(-\frac{1}{2}\right)^n.$$

Hence

$$\boxed{L_n = \frac{700000}{3} + \frac{800000}{3} \left(-\frac{1}{2}\right)^n.}$$